CAREFLOW

Pandas Work — Team Handover Guide

Important data note: The supplied dataset is synthetic. Source timestamps contain clock times only, not full calendar dates. Journey durations and transition gaps derived from reconstructed timestamps are provisional until validated.

1. Project and ownership

CareFlow explores patient activity sequences using Python/Pandas, with MySQL for storage, PM4Py for process mining, and Power BI for dashboard visualizations. The Pandas contribution covers dataset inspection, quality checks, repeated-activity analysis, journey-duration and transition-gap calculations, and preparing outputs for teammates.

2. Folder structure (based on the VS Code screenshot)

This reflects the visible workspace: the source Excel files and notebook are inside **Care Flow/python scripts/**, and the export folder is named **output of pandas**. The reference-pathway filename below follows the screenshot spelling; use the exact filename present in your Explorer.

```
Care Flow/
■■■ python scripts/
    ■■■ cleaning.ipynb
    ■■■ care_flow_case_details.xlsx
    ■■■ care_flow_event_log.xlsx
    ■■■ careflow_reference_pathway.xlsx
    ■■■ output of pandas/
    ■   ■■■ careflow_cleaned_events.xlsx
    ■   ■■■ careflow_journey_summary.xlsx
    ■   ■■■ careflow_repeated_activities.xlsx
    ■   ■■■ careflow_transition_analysis.xlsx
    ■   ■■■ careflow_patient_summary.xlsx
    ■■■ CareFlow_Dataset_Understanding_Document...
    ■■■ CareFlow.docx
    ■■■ README.md
```

The output files are expected only after their corresponding export cells have been run successfully. The screenshot does not confirm that every output file exists yet. The folder tree intentionally does not place the handover PDF at a location that has not been verified.

3. Input files

care_flow_event_log.xlsx — event-level table; 10,485 rows and 8 columns. Columns: Case_ID, Activity_Name, Timestamp, Event_Order, Department, Triage_Priority, Shift, Doctor_ID. Used for activity sequences, activity counts, repeated-activity checks, and transition gaps.

care_flow_case_details.xlsx — case-level table; 1,500 cases and 9 columns. Includes Case_ID, Age, Sex, Triage_Priority, Department, Shift, Assigned_Doctor_ID, Arrival_Timestamp, and Discharge_Status. Used to segment journey results by patient/case attributes.

careflow_reference_pathway.xlsx — illustrative reference-pathway steps. Intended for sequence comparison; these templates are not clinically validated medical guidelines.

4. Notebook workflow: cleaning.ipynb

Task	Methods	Purpose
Load data	pd.read_excel()	Load Excel files into DataFrames.
Inspect structure	head(), shape, columns, info()	Understand rows, columns, and data types.
Check quality	isnull().sum(), duplicated().sum()	Check missing values and exact duplicates.
Count categories	nunique(), value_counts()	Count cases and activity frequencies.
Repeated activities	groupby().size()	Find activity types repeated within cases.
Journey summary	groupby().agg()	Summarize reconstructed start/end times and duration.
Transition table	groupby().shift() + time subtraction	Calculate gaps between consecutive events.
Merge case information	merge()	Combine case attributes with calculated metrics.
Export results	to_excel()	Save analysis tables for teammates.

5. Initial data-quality results

Observed in the notebook: 10,485 event rows; 1,500 unique cases; 8 activity types; zero missing values in event-log columns; zero exact duplicate rows; Event_Order increasing for all 1,500 cases.

Activity	Event count
Doctor Consultation	2,816
Triage	1,788
Registration	1,500
Treatment	1,500
Discharge	1,500
Blood Test	659
X-Ray	657
CT Scan	65

An initial analysis run reported 1,158 cases with repeated activities (77.20%). Re-run and validate the notebook before treating this as final.

6. Output files and what they support

careflow_cleaned_events.xlsx — event rows plus derived columns such as reconstructed timestamp, previous activity/time, and transition gap (exact columns depend on cells run). Supports Total Patients, Total Events, activity frequencies, repeated-activity analysis, and transition-gap analysis.

careflow_journey_summary.xlsx — one row per Case_ID with Start, End, and Duration_Min. Supports Average Journey Duration and Median Journey Duration. Durations are provisional until timestamp assumptions are validated.

careflow_repeated_activities.xlsx — case/activity combinations where an activity occurs more than once, with Count. Supports repeat investigation and Repeat Activity Rate = cases with at least one repeated activity ÷ total cases × 100.

careflow_transition_analysis.xlsx — one row per Previous_Activity → Activity_Name transition, with count, mean, and median gap. Supports comparison of recorded transition gaps; a gap is not automatically verified queue waiting time.

careflow_patient_summary.xlsx — intended merge of case details, journey duration, and repeat fields. Supports comparisons by triage priority, department, shift, age group, and discharge status. Verify that the merge/export ran and there is one row per Case_ID.

7. KPI mapping

KPI	Definition	Main file(s)
Total Patients	Distinct Case_ID count.	cleaned events / source event log
Average Journey Duration	Mean per-case duration in minutes.	journey summary
Median Journey Duration	Median per-case duration in minutes.	journey summary
Average Transition Gap	Mean gap between consecutive event timestamps, in minutes.	cleaned events / transition analysis
Repeat Activity Rate	Cases with any repeated activity ÷ total cases × 100.	repeated activities + event log
Pathway Deviation Rate	Percentage of cases differing from a reference pathway under a documented rule; not yet implemented in described code.	event log + reference pathway

8. Timestamp limitation

Timestamp and Arrival_Timestamp contain times only, with no dates. Direct conversion may attach a default date such as 1900-01-01, which is not a real event date. The notebook explored constructing a relative timeline using arrival time, Event_Order, and midnight rollover. This is an estimate based on assumptions, not recovery of verified timestamps. Validate before publishing journey-duration or transition-gap KPIs.

9. Handover to teammates

• **MySQL:** use validated event/case tables and keep Case_ID consistent as the join key.
• **PM4Py:** use Case_ID as case identifier, Activity_Name as activity, and a validated chronological timestamp.
• **Power BI:** use agreed and validated patient/journey/transition tables; do not label timestamp gaps as confirmed queue waiting times.
• **All teammates:** the dataset is synthetic; findings demonstrate the workflow, not actual hospital performance.

10. Before final submission

■ Restart kernel and run all cells in order.
■ Confirm expected Excel outputs exist.
■ Check output row counts and nulls after merges.
■ Verify patient summary has one row per Case_ID.
■ Validate timestamp reconstruction before publishing duration KPIs.
■ Implement and document the pathway-deviation rule before reporting that KPI.